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Professional Summary

Senior Data Scientist with 7 years of experience leading data-driven transformation initiatives through machine learning, advanced analytics, and business intelligence. Skilled in building production-ready ML solutions, developing scalable data products with Python and SQL, and communicating insights through Power BI and Tableau. Experienced in managing end-to-end project lifecycles, collaborating with cross-functional stakeholders, and delivering measurable outcomes across diverse business domains at Gramener, and ValueMomentum and Innodatatics.

Technical Skills:

Programming & Analytics	Python, R,SQL
Machine Learning	Scikit-learn, XGBoost, LightGBM, Statsmodels, Supervised & Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Ensemble Methods, Model Evaluation & Validation
Deep Learning	PyTorch, TensorFlow, Keras, CNNs, LSTMs, Transformers
NLP	NLTK, spaCy, BERT, Text Classification, Named Entity Recognition (NER), Sentiment Analysis
Computer Vision	OpenCV, YOLO, VGG19, Image Classification, Object Detection, Image Processing
Big Data & Data Engineering	PySpark, ETL Pipelines, Data Processing & Transformation
Data Visualization & BI	Power BI, Tableau, Matplotlib, Seaborn, Dashboard Development, Data Storytelling
Application Development & Automation	Flask, Streamlit, REST APIs, Selenium, BeautifulSoup, Web Scraping
Databases	Oracle, PostgreSQL, InfluxDB
Cloud & MLOps	AWS, Azure, Docker, Kubernetes, Model Deployment, Containerization

Executive Summary:

- Senior Data Scientist with 6+ years of experience delivering data-driven solutions across Finance, Insurance, and Manufacturing domains.
- Expertise spanning Business Intelligence, Advanced Analytics, Machine Learning, and MLOps, enabling end-to-end solution ownership.
- Skilled in Python, SQL, and PySpark for building scalable data pipelines, automation frameworks, and production-grade ML solutions.
- Led client-facing Proofs of Concept (POCs) and transformed complex business challenges into actionable, production-ready solutions.
- Strong track record of driving stakeholder adoption through effective communication, technical leadership, and business-focused problem solving.

Employment History:

🚦 Senior Data Scientist (Data Science & AI), Gramener,Bangalore, Karnataka (Oct 2023 - Present).

Live Projects done:

Rule Variable Integrity Analyzer – JSON Audit Automation

Client Project | Finance Domain (Fraud/Risk Analytics)

Role: Senior Data Scientist

Objective:

Developed a Python-based automation tool to assess the reliability and completeness of variables used in rule-based fraud detection and risk analysis systems. The goal was to validate field fill rates and computation accuracy from complex, nested JSON data structures to ensure sound decision-making.

Key Responsibilities:

- Designed a dynamic parser to extract, analyze, and validate variables from diverse JSON formats, including nested objects, stringified arrays, and deeply embedded lists
- Implemented logic to automatically generate access codes (dot notation or index-based extraction) for complex fields using Python
- Created rule audits that flagged low fill-rate or malformed fields for review by analysts and fraud strategy teams
- Standardized field validation across multiple datasets with varying JSON schemas
- Collaborated with fraud analytics teams to improve upstream data engineering pipelines by identifying broken or miscomputed fields

Tools & Tech Stack:

Python, pandas, JSON, Regex, Jupyter, VSCode

Impact:

- Automated analysis reduced manual variable review time by >80%
- Flagged 100+ critical variable issues, improving fraud model reliability and rule quality
- Enabled seamless integration with Hive by storing access codes in structured tables, eliminating the need for manual SQL checks to verify variable presence and usage

Fraud Detection Model Monitoring & Alerting Framework**Client Project | Finance Domain (Fraud/Risk Analytics)****Role: Senior Data Scientist****Objective:**

Built a monitoring and alerting framework for payment fraud detection models to proactively identify model degradation, data quality issues, and scoring anomalies in production.

Key Responsibilities:

- Developed model and data quality alarms using Python, SQL, and PySpark for fraud detection models.
- Built monitoring dashboards and automated alerting mechanisms for production model health tracking.
- Optimized monitoring architecture by consolidating version-specific datasets into a scalable framework.
- Implemented near real-time monitoring using Kafka, reducing latency from 1 hour to 10 minutes.

Tools & Tech Stack:

Python, SQL, PySpark, Hive, Kafka, Powerbi

Impact:

- Improved visibility into fraud model performance and data quality.
- Enabled faster detection of production anomalies through automated monitoring.

VRAS Surgical Analytics Dashboard**Client Project | Healthcare Analytics (Clinical Trails)****Role: Power BI Developer****Objective:**

Develop an analytics dashboard for the VELYS™ Robotic-Assisted Surgery platform to provide visibility into surgical performance, workflow efficiency, patient outcomes, and procedure-level metrics for clinical and operational stakeholders.

Key Responsibilities:

- Designed and developed Power BI dashboards to monitor robotic-assisted surgical procedures, workflow performance, and operational metrics.
- Built analytical views for surgical profiles, procedure volumes, workflow durations, case-level details, and facility performance.
- Developed visualizations for planned vs. actual surgical outcomes, implant positioning, alignment measurements, and soft tissue balancing metrics.

- Created scalable data models and DAX calculations to support clinical performance analysis and reporting.
- Collaborated with stakeholders to translate business requirements into interactive dashboards and actionable insights.

Tools & Tech Stack:

Power BI, DAX, SQL, Power Query, Data Modeling

Impact:

- Delivered centralized visibility into robotic-assisted surgical procedures across facilities and surgeons.
- Improved monitoring of surgical workflows, procedure efficiency, and clinical outcome metrics.

🌈 **Data Scientist (Data Science & AI), Value Momentum, Hyderabad, Telangana (Jan 2022 – Oct 2023).**

Early Claims Prediction

Client Project | Insurance Domain (Predictive Analytics)

Role: Data Scientist

Objective:

Develop a predictive model to identify insurance claims likely to be filed early, enabling proactive risk assessment, improved claims management, and optimized business operations

Key Responsibilities:

- Performed exploratory data analysis and feature engineering on policyholder, claims, and transactional data.
- Built and evaluated machine learning models to predict the likelihood of early claim occurrences.
- Applied feature selection and model tuning techniques to improve predictive performance and model interpretability.
- Collaborated with business stakeholders to translate analytical findings into actionable insights.
- Developed dashboards and reports to communicate model performance and key risk drivers.

Tools & Tech Stack:

Python, SQL, Scikit-learn, XGBoost, Pandas, NumPy, Power BI

Impact:

- Enabled early identification of high-risk policies for proactive intervention.
- Supported business teams in optimizing claims management and resource allocation.

AI Cognitive Solution for Financial Document Extraction and Aadhar Masking

Client Project | Insurance Domain (Computer Vision)

Role: Data Scientist

Objective:

Reduce manual effort in document review by automating Aadhaar extraction and masking to support privacy and regulatory compliance

Key Responsibilities:

- Developed a Computer Vision solution for automated Aadhaar detection, extraction, and masking from documents.
- Built YOLOv5 and VGG19 models for document identification, rotation correction, and masking workflows.
- Integrated OCR (Tesseract, EasyOCR) for extracting Aadhaar information from scanned documents.
- Developed REST APIs and stored extracted data in Oracle DB for downstream consumption.

Tools & Tech Stack:

Python, YOLOv5, VGG19, OpenCV, TensorFlow, Keras, Tesseract OCR, EasyOCR, Oracle DB, REST APIs

Impact:

- Automated document processing and Aadhaar masking, significantly reducing manual effort.
- Improved data privacy compliance through consistent handling of sensitive customer information.

🌈 **Associate Data Scientist (Data Science & Machine Learning), Innodatatics*Innovation | Data | Analytics,** Hyderabad, Telangana (August 2019 – Jan2022).

Predictive Maintenance for Manufacturing Equipment

Client Project | Manufacturing (Food & Beverage)

Role: ML Engineer

Objective:

Develop predictive models using machine sensor data to forecast machine failures and minimize production downtime in beverage manufacturing operations.

Key Responsibilities:

- Extracted and processed sensor data from InfluxDB for predictive maintenance analysis.
- Performed EDA and automated data profiling using SweetViz, Bokeh, Pandas, and Python.
- Developed forecasting and survival analysis models using Kaplan-Meier, ARIMA, and SARIMA techniques.
- Built Tableau dashboards to visualize machine health, failure trends, and predictive insights.

Tools & Tech Stack:

Python, Pandas, NumPy, Scikit-learn, Statsmodels, InfluxDB, Tableau, Flask, Bokeh, SweetViz

Impact:

- Enabled proactive identification of potential machine failures using sensor-based predictive analytics.
- Reduced unplanned downtime and supported maintenance planning through data-driven insights.

Stock Market Sentiment Analysis using BERT

Client Project | Financial Analytics

Role: Associate Data Scientist

Objective:

Analyze financial news sentiment and evaluate its relationship with stock price movements using NLP and transformer-based models.

Key Responsibilities:

- Collected and processed financial news headlines from Finviz using BeautifulSoup and Selenium.
- Performed text preprocessing and sentiment analysis using NLP techniques and traditional models such as VADER, TextBlob, and RNN.
- Implemented and evaluated BERT-based sentiment classification models using transformer architectures.
- Developed and deployed an interactive sentiment analysis application using Streamlit.

Tools & Tech Stack:

Python, BERT, Transformers, TensorFlow, NLTK, BeautifulSoup, Selenium, Streamlit, Heroku, Matplotlib, Seaborn

Impact:

- Enabled automated sentiment analysis of financial news to support market trend assessment.
- Demonstrated the relationship between news sentiment and stock price movements using NLP-driven insights.

Educational Qualification:

Course	Specialization	Institute	Year	Percentage
B. TECH	Computer Science	Kakinada Institute of Engineering and Technology	(2015-2019)	66.55
Senior Secondary	-	Sri Chaitanya Junior college	(2013-2015)	85.50
Higher Secondary	-	New Cambridg High School	2013	80.75

Certifications and Training:

- a. **“Data Science using Python”** certification course by **360digitmg**
- b. **“Introduction to Artificial Intelligence”** online certification course by **360digitmg**
- c. **“Tableau”** certification course by **360digitmg**

Achievements and Recognition:

- **Technology Champion Award** – Recognized for outstanding technical contributions and delivery excellence across TikTok projects at Straive (2024–2026).
- **Global Azure Speaker Award** – Recognized for delivering technical sessions and knowledge-sharing initiatives within the Azure community (2023)

Declaration:

I do hereby declare that all the information provided above are true to the best of my knowledge and belief.

Sai Sandeep K